Supporting information: recovery, screening and trend inference

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Contents

The following sections document source-comparison denominators, whole-archive and interval procedures, reference qualification, and sensitivity tables for the main article. Source data, model outputs and software require the separate access arrangements described in the article's Open Research section; this file does not substitute for those materials.

S1 Design and information separation

The source comparison uses twenty repetitions of masks and group-role assignment on the same assembled archive; these are not independent climates. Held test connected groups are excluded from estimator fitting, interval calibration, risk learning and rule calibration. In the original partitioned forward models, other training-only groups in a named target region can remain in source roles. The source-role audit and subsequent strict-region sensitivities distinguish this exception from the held target-group exclusion.

In each whole-archive setting, selected gaps in the target station are masked together before local daily fitting. Unmasked target observations and the designated donor observations remain available. The source annual model is fitted separately and held fixed within that archive reanalysis. The first seven years are observed burn-in; the forward evaluation masks only years from 2010 onward. Twenty repetitions at the same station are not twenty independent rivers. The heat-associated and synchronized-donor stresses have small station samples; the former also changes the selected missing years and their number.

S2 Scoring and interval construction

For ensemble $x_1, \ldots, x_M$ and outcome y, the continuous ranked probability score is

$$M^{-1} \sum_m |x_m - y| - (2M^2)^{-1} \sum_{m,l} |x_m - x_l|.$$

Threshold-weighted scoring replaces samples and outcome by their maxima with the training H30 threshold. The 90% interval score adds twenty times the distance outside the interval to its width. A 95% trend interval score uses the corresponding factor forty. Missing predictions remain unavailable rather than being assigned zero error.

Both annual estimators use the observed-window support projection where a finite bound exists. Seven stored quantiles do not determine their outer tails; the normal-spacing continuation and the alternative constant tail are assumptions. Cross-year rank constructions preserve annual marginal samples but are not a posterior for the fitted annual model. The source-only corrections and the post-hoc propagation reported below do not establish a finite-sample or independent-climate coverage guarantee.

For the exploratory release-then-trend analysis, an observed lower bound L contributes $\log \Phi\{(\mu - L)/\sigma\}$ to a Gaussian working likelihood. Released numerical estimates are treated as exact, unknown cases remain unavailable, and the working interval includes regression variability. It is not a calibrated missingness-only interval.

S3 Core source comparisons

Table S1. Dangerous source-case summaries: the gap overlaps at least half the training-period peak window. Entries are medians across 20 repetitions sharing the source archive. Method-specific availability can differ; paired case comparisons are reported separately. Coverage and CRPS are not defined for point-only methods.

Family	Method	Availability	MAE	90% coverage	CRPS
calendar	D2	100.0%	0.451	86.7%	0.329
calendar	functional	100.0%	0.476	–	–
calendar	plug-in	100.0%	0.564	–	–
calendar	classical	99.9%	0.463	86.0%	0.336
crossed	D2	100.0%	0.337	89.3%	0.246
crossed	functional	100.0%	0.355	–	–
crossed	plug-in	100.0%	0.428	–	–
crossed	classical	99.7%	0.336	89.8%	0.245

Table S2. Held-region source-calibrated comparison of the three-output and score-only rules. The score cutoff is chosen from disjoint source rule-calibration groups. Retention differs in the evaluated target cases; errors are conditional on numerical publication.

Family	Method	Eligible	Published	Retention	Error rate
calendar	score only	31,672	26,896	84.9%	0.0%
calendar	three outputs	31,672	27,090	85.5%	0.7%
crossed	score only	114,012	80,980	71.0%	3.9%
crossed	three outputs	114,012	82,169	72.1%	4.9%

Table S3. Retrospective equal numerical-count comparison. The score threshold uses the evaluated feature distribution and the three-output count, so this is transductive. Equal numerical counts do not equate retained bounds or unknown states.

Condition	Family	Eligible	Published by each	Three failures	Score failures	Three / score error
air substitution	calendar	31,672	88	19	23	21.6% / 26.1%
clean	calendar	31,672	27,090	190	4	0.7% / 0.0%
clean	crossed	114,012	82,169	4026	3249	4.9% / 4.0%
future	crossed	27,846	17,583	1924	1220	10.9% / 6.9%

Table S4. Annual and trend coverage on the common archive cohort. Annual coverage weights cases by incomplete years. Tail and rank experiments use different random draws; the three rank methods share the same annual marginals. Repetitions share the same source archive.

Family	Mode	Method	Cases	Groups	Annual 90%	Constant 95%	Normal 95%	Independent 95%	Shared 95%	Blocks 95%
calendar	forward	D2	818	17	98.2%	94.6%	95.7%	96.0%	96.1%	96.3%
calendar	forward	classical	699	17	98.2%	94.6%	95.3%	95.4%	95.4%	95.7%
calendar	offline	D2	840	18	98.5%	92.3%	94.9%	95.0%	95.1%	94.9%
calendar	offline	classical	835	18	97.4%	91.7%	93.3%	93.2%	93.3%	93.3%
crossed	forward	D2	818	17	90.1%	83.0%	88.8%	88.0%	87.9%	89.0%
crossed	forward	classical	736	17	88.1%	81.4%	86.3%	86.3%	86.1%	87.2%
crossed	offline	D2	840	18	94.1%	91.7%	94.2%	94.0%	94.5%	94.8%
crossed	offline	classical	838	18	92.6%	86.4%	90.7%	90.2%	91.6%	91.2%

S4 Whole-archive supplements

Full paired MSE cells, group and region sensitivities, unavailable cases and source calibration support are in the review-response results. Forward task calibration ends before 2010. Normal-spacing tail extrapolation is compared to constant tails under the same observed-window constraint; its outer distribution is assumed, not identified by the seven stored quantiles. Auxiliary rank coupling is not a posterior for the annual estimator.

S5 Matched numerical release within each archive

After the original release and trend results were inspected, we fixed a further descriptive comparison on the same 3,320 archive settings. In each archive the three-output rule's number of numerical releases was matched exactly by choosing the lowest available raw classifier scores; equal scores were broken by year, and the existing detected-quality exclusion was applied to both rules. The choice uses the target archive's score distribution and its original release count, but no reference outcomes. It is therefore transductive and not a prospective release threshold. The same observed years and D2 estimates are used, while the selected years can differ. Nonselected years retain a visible lower bound if allowed by the same quality flag, and otherwise remain unknown. An unassessed quality diagnostic does not count as a detected flag in this analysis, as in the original pipeline.

Table S11 gives annual errors and working-likelihood trend RMSE; lower-bound counts are not an assertion that the same years or bound values were retained. The number of numerical outputs is matched within each archive, but equal counts do not establish equal total information or comparable prospective risks. Repeated masks are not independent climates. A connected-group bootstrap of paired squared-error differences for forward controlled gaps gives [-0.0017, 0.0072] $(^\circ\mathrm{C/decade})^2$; this is descriptive and spans zero. The likelihood treats released predictions as exact, censoring may be informative, and its intervals are not calibrated missingness-only uncertainty. No model was refitted for this comparison.

The audit separately reconstructed the selection masks for all 3,320 archives using Python ordering, recomputed 3,319 available score-comparator numerical-only OLS slopes and checked all eight summary cells. A second implementation refitted

Table S5. The same 35 observation-only complete-year witnesses under each screening rule. The accepted-record comparison uses observed windows, not unknown complete H30 values. The 49-day and July–August rows are contextual proxies, not published H30 rules.

Rule	Witnesses	Admitted	Excluded	Above accepted
Complete	35	0	35	35
Annual 95	35	31	4	3
Station warm 90	35	32	3	2
Fixed warm 90	35	32	3	2
July–August	35	30	5	3
Gap ≤49 d	35	34	1	1

Table S6. Seven-station stress comparisons on common available station/repetition rows. Heat-associated masks and numbers of missing years differ between arms; unavailable cases remain in the eligible denominator. Coverage and width are supplied in the result table.

Scenario	Eligible	Available	Base RMSE	Stress RMSE
Donors / forward	30	30	0.079	0.116
Donors / offline	35	35	0.037	0.047
Heat / forward	30	30	0.079	0.088
Heat / offline	35	30	0.035	0.058

Table S7. Release followed by finite-record trend inference for controlled gaps. The values-and-bounds interval is a Gaussian working-likelihood interval, including regression variability; it is not calibrated missing-data uncertainty.

Mode	Released data	Eligible	Available	RMSE	Working width
forward	Values + bounds	820	820	0.202	3.262
forward	Values	820	820	0.210	–
offline	Values + bounds	840	840	0.040	1.986
offline	Values	840	840	0.043	–

Table S8. Task calibration and independent-draw intervals on exactly the same supported cases. Coverage is a fraction; missing source support is reported separately.

Setting	Method	n	Original coverage	Task coverage	Width ratio
calendar / forward	D2	40	0.925	0.975	0.947
calendar / forward	Stochastic	38	1.000	1.000	0.773
calendar / offline	D2	95	0.937	0.958	1.167
calendar / offline	Stochastic	88	0.977	0.989	1.225
crossed / forward	D2	39	1.000	1.000	1.112
crossed / forward	Stochastic	38	1.000	1.000	1.015
crossed / offline	D2	301	0.977	0.980	1.236
crossed / offline	Stochastic	307	0.948	0.951	1.108

Table S9. Task calibration support. Empirical calibration from small source pools does not establish a finite-sample guarantee.

Family	Time setting	Estimator	Availability	Minimum source groups
calendar	forward	D2	0.049	3
calendar	forward	stochastic_calibrated	0.046	3
calendar	offline	D2	0.113	3
calendar	offline	stochastic_calibrated	0.105	3
crossed	forward	D2	0.048	3
crossed	forward	stochastic_calibrated	0.046	3
crossed	offline	D2	0.358	3
crossed	offline	stochastic_calibrated	0.365	3

all 6,640 rule-by-archive censored likelihoods using normal log-density and survival functions. Convergence status agreed for every fit; 6,638 converged in both implementations, and their largest slope difference was approximately 1.2×10^{-6} °C per decade. This numerical check used the same observations, selected cases and working likelihood; it did not independently reconstruct source sensors or models, establish likelihood validity or verify interval standard errors.

S6 Expanded missingness-mechanism sensitivities

After the earlier results and seven-site pilot had been inspected, we fixed an expansion to the first five masks for all 41 forward-eligible original archive stations in 17 connected groups. Only controlled gaps and one seed-zero source-model assignment are studied. For the donor sensitivity, the target masks are unchanged and those same dates are hidden in every other panel column before local fitting. All 205 settings have at least one finite donor-panel observation removed; the total of 427,375 finite donor-panel cells counts repeated masks and may include columns not ultimately selected by the local estimator. It is not a count of independent outages or necessarily of observations the estimator would have used.

For the heat-associated sensitivity, the original 35% Bernoulli draw determines the number of masked years in each station–repetition archive. An independent seeded exponential-race sample chooses exactly that number of eligible years, with weight five for years whose annual air H30 is at least the first-seven-year 80th percentile, and weight one otherwise. Air information determines a synthetic test mask; it is not a hidden water-temperature outcome. The full cohort has 943 masked years in either design, of which 511 and 700 are classified as air-hot. However, the selected years have different existing within-year gap shapes and positions, so the total hidden target days change from 55,950 to 57,540. These paired scenarios cannot isolate the causal effect of heat association alone or represent a measured outage mechanism.

Table S12 compares fixed-prediction cohorts for both estimators, with availability reported on the common subset. Nominal trend interval coverage, widths and scores use independent annual ranks and the original normal-spacing tails; the intervals are not validated climate-transfer guarantees. Connected-group resampling

of paired D2 squared-error differences gives [0.0030, 0.0434] for synchronized donors and [-0.0040, 0.0091] for heat-associated masking, in $(°C/decade)^2$. These describe a selected archive and depend on group exchangeability. Removing Bavaria without refitting changes baseline and donor RMSE to 0.0829 and 0.1287, and baseline and heat RMSE to 0.0829 and 0.0816. Other fixed-prediction leave-region-out summaries are supplied in the aggregate files. Repeated masks are not independent climates; original forward source roles are not uniformly strict named-region holdouts.

An independent audit reconstructs all 205 baseline masks, 205 synchronized masks and 205 heat-associated masks, and recomputes eight paired RMSE and mean-squared-error cells. It uses the same source observations, random seeds and saved fitted predictions; it does not independently refit the models, check bootstrap intervals or establish sensor quality.

S7 Source-model sensitivity of expanded stresses

Following inspection of the original source-role audit and expanded stress results, we replayed the stressed daily features and masks through two already fitted alternative source-model sets. The strict-original version replaces only the forward HUC17 model, the sole affected target region in this cohort, with the model refitted after excluding every HUC17 connected group from its source roles; other source assignments remain as before. This removes source cases and changes source sample size. The candidate version replaces the models for all six forward target regions with the separately refitted candidate-provider models, which exclude all target-region groups but also change source providers, source sample size and role assignments. Source-model outputs and calibration offsets are recalculated; the previously fitted local daily features and target masks remain fixed. Neither comparison is prospective or a clean causal experiment for a single change.

Table S13 uses, separately for each estimator and stress, only archives available under both baseline and stress in all three source versions. The independent annual-rank intervals use the same six-way common interval cohort, with the same normal-spacing tail assumptions. D2 has 205 common archives; the stochastic method has 190 for donor gaps and 187 for the fixed-count heat stress. Repeated masks are not independent climates. Descriptive connected-group bootstrap and

fixed-prediction region deletion results accompany the aggregate files. In the heat scenario the number of masked years is fixed within archive, but the number and placement of hidden days can change; in the donor scenario target masks are identical. These simulations do not verify a natural outage mechanism or transfer to new climates.

The source-role audit checked the positive-weight roles of the original and alternative models. All 440 alternative stress reanalyses preserved the target masks, hidden-value checks, plug-in values and observed lower bounds. The audit checked all 205 candidate baseline receipts against their source models, calibration, original features and saved outputs, and replayed the 15 affected strict-original baselines, including 60 annual predictions and 180 trend rows. A second implementation independently recomputed the six-way common cohorts, 24 aggregate cells, 144 region-deletion cells and 4,000 group bootstrap draws per cell from 9,840 saved case rows. These checks did not independently refit source models, check sensor accuracy or validate climate transfer.

S8 Exploratory projected-point failure score

After inspecting prior projection and risk results, we fitted six region-specific binary classifiers to identify absolute D2 point error above 0.5 °C after the observed-bound projection. For each target region, the source estimator, interval-calibration, risk-learning and rule-calibration connected-group roles are disjoint, all groups appearing in the target region are excluded, and forward source cases precede 2010. Training uses the assigned risk-learning source groups, weighted equally by connected group. The new classifier uses the existing decision-time feature matrix, the projected 90% interval width and the projected point's distance above the visible bound; neither the old risk score nor the held reference is a predictor. Each model uses 120 low-leaf-count boosted trees with the fixed settings documented in the analysis plan. Six to eight source training groups contribute per model.

For each region and mask family, separate source rule-calibration groups select an old-score and a new-score cutoff to approximate 80% group-equal-weight numerical retention on valid source cases, without using failure labels to choose the cutoff. Both cutoffs are fixed before target scoring. Target numerical retention can

differ. A second, transductive comparison ranks the new target scores and matches the old-score numerical count separately within each target region and family; ties are broken deterministically without target outcomes. It is not a prospective cutoff, and equal counts need not retain the same years or bound values.

The candidate annual caches contained no reference H30 values. Therefore source models, source cutoffs, target scores and selections were saved first; the target reference was then joined by station, masked year, family and replicate from the separately saved complete-archive record. A second comparison against the original source station–year–family H30 verified all 7,583 reference values, and the saved selections did not change. These target archives and source data had nevertheless been inspected in earlier analyses; this is not an unopened reference experiment.

Table S14 gives pooled forward results. Source thresholds in the controlled-gap family give old/new numerical retention 89.6% / 85.3% and selected failure 10.72% / 9.50%. At the matched 3,385 outputs, the original and projected-risk scores have 363 and 383 failures, respectively; the connected-group bootstrap interval for the difference in selected failure rates (new minus old) is [-0.73, 1.86] percentage points. Group resampling is descriptive and repeated masks are not independent climates. The old classifier targeted an unprojected point; the new classifier targets the projected point, but a better-aligned training label does not itself imply better deployment decisions. The source masks degrade one year at a time, while target archives are jointly degraded across years. These results do not establish calibrated probabilities, group-level risk control, natural-outage performance or independent-climate transfer.

The independent audit refits the six classifiers using the same source records and feature implementation, rechecks the disjoint roles, reproduces the twelve family-specific source threshold pairs, reconstructs all target scores and label-free selections, and recomputes group bootstrap and region-deletion summaries. It also checks reference attachment against both saved reference sources. It does not independently reconstruct physical observations, establish source permissions or verify the classifier's prospective reliability.

S9 Seasonal-window and screening-rule diagnostics

The previously selected source sample contains 96 eligible long-record stations; this calculation uses the 95-station project-processed subset with a documented candidate redistribution path. None of the 35 recorded witnesses is at the excluded LUBW station, although it remains part of the original eligible-station denominator. The subset comprises 2,122 complete station-years at 95 stations and 33 connected groups. For each complete year, we compare the full-year reported H30 with the largest 30-day mean whose every day lies from 1 July through 31 August. Of those years, 779 exceed the summer-window value by more than the numerical tolerance 5×10^{-6} °C, and 304 exceed it by more than 0.5 °C. The mean difference is 0.255 °C. These are complete recorded years, not simulated reconstruction errors; neither the tolerance nor the 0.5 °C threshold is an estimate of sensor error.

The comparison was specified after the original witnesses and literature had been examined. It is not a reproduction of Beaufort et al.'s multiyear thermal peak or their summer air-temperature climate correction; that study also excluded sites with strong human influence. The present sample contains regulated and incompletely characterized sites, covers different locations and years, and the fully-contained July–August window differs from a peak-center criterion. Consequently our fraction must not be compared directly with their reported fraction. The incomplete-year comparison is also conditional on the 35 already selected record witnesses: all have a fully observed July–August 30-day window, and 9 have a larger fully observed 30-day window elsewhere in the year. This does not establish the full H30 of an incomplete year or a population prevalence.

Table S15 counts previously eligible station-years rejected by four explicit study rules, including how many still supply at least one fully observed 30-day window. A lower bound is information about the reported record, not a complete annual estimate or a guarantee of better trends. The 90% station-specific warm-season rule uses a leave-target-year-out climatology; the complete July–August row is a contextual comparator rather than the full published climate-correction procedure. The records are not an independent validation sample and no sensor error has been estimated.

After inspecting the largest annual gaps, we separately summarized existing project regulation labels and removed the two sites with the largest individual gaps (Table S16). The 2010 USGS analysis identified Blue River and Fall Creek as dam-affected monitoring sites in 2001–2002; it does not establish the cause or measurement quality of every year examined here. The existing *unmatched gages* label does not mean unregulated, and labels and sites were not randomized. Removing these two post-hoc sites leaves 749 of 2,086 complete station-years above the numerical tolerance, with mean difference 0.197 °C. These summaries do not identify causal dam effects or transfer to streams absent from the sample.

The comparison and screening totals were independently reconstructed using pandas rolling windows on the same snapshot. The post-hoc influence totals and 36 selected-site peak-window dates were independently checked using NumPy convolution; the stored regulation labels were checked against the project's source metadata. Neither audit verifies the original provider's measurements or whether each project regulation label is accurate. The supplied 95-station snapshot permits the window and screening arithmetic to be rerun, subject to its attribution and reuse conditions.

S10 Past-only warm-season sensitivity

The original station-specific warm season leaves out the evaluated year but can use observations from later years. After inspecting that comparison and the 35 selected witnesses, we recomputed the same 15-day-smoothed day-of-year climatology and five-month selection with only daily values from years strictly before each evaluated year. The source is the current processed archive, not a historically timestamped data vintage. The 95 stations and 35 witnesses were themselves selected using records available after some evaluated years, so this is a retrospective information-separation sensitivity, not a prospective site recruitment or field performance test.

Two past-only variants are reported. One uses any earlier finite water observation, which can amount to only 1 day for an incomplete year; climatological interpolation with so little data is not evidence of adequate support. The other requires at least four fully observed prior calendar years before classifying an in-

complete year. Complete years are accepted even if the climatology is unavailable, because every possible warm season is fully observed. The four-year threshold is an exploratory study choice, not a published standard. Unsupported incomplete years remain unknown rather than being counted as rejections. On 2,656 eligible station-years, the past-only variant without a minimum changes 1 decision relative to the original leave-year-out rule; the four-year version leaves 127 years unknown, 122 of which still have an observed 30-day lower bound.

Table S17 also compares the reported observed-window maximum in each rejected year to the greatest reported window from earlier years accepted by that same rule. A rejected year contributes only if both its observed bound and an earlier accepted reported window exist. Earlier accepted incomplete years can have an unobserved higher H30, and these are not comparisons with physical sensor truth. The leave-year-out rule's earlier classifications may use years later than their evaluation year; only the past-only rules enforce the stated temporal information boundary. Among the 35 retrospectively selected witnesses, the four-prior-complete-years variant accepts 27, rejects 2 and leaves 6 unknown; 2 rejected witnesses exceed the earlier accepted reported-window maximum. These counts are not an incidence estimate.

An independent implementation rebuilt all 2,656 earlier-only climatologies, 13,280 rule states and prior-record comparisons on the same processed daily records. In 35 witness-year probes, shifting the target and all later observations by 5 °C did not change the earlier-only warm season. These checks do not verify historical data vintages, station selection, provider quality control or physical measurement accuracy.

S11 Conservative source-group maximum sensitivity

After earlier annual, trend and classifier results had been inspected, we combined the candidate source model's assigned risk-learning and rule-calibration connected groups for each target region. These groups are disjoint from the numeric quantile estimator and its interval-offset groups and exclude groups in the held target region. The risk-learning groups trained a separate failure classifier, but no classifier score or selection is used in this interval analysis. Replacing the stored

classifier with a constant score in 300 source-case probes left both methods' seven annual quantiles unchanged. Source cases are before 2010 and use single-year masks. The source and target observations were already opened; this is a post-hoc sensitivity, not a new prospective experiment.

For each valid source group, we calculate the maximum of the nonnegative distances by which a reference falls outside its projected 90% interval. With G groups, the expansion is the $\lceil 0.9(G + 1) \rceil$-th ordered group maximum if that rank exists and is positive; otherwise expansion is unsupported. The six forward regions have 12–16 contributing groups per method and family, so the chosen rank is the maximum among the available group maxima. This finite order statistic does not establish a 90% group-coverage guarantee: exchangeability is not demonstrated, source cases use individual-year masks, target archives have joint multiyear gaps, and groups have different sizes and climates. Both methods' lower endpoints are constrained to the visible observed window. Table S18 compares the same target cases; the hidden-peak subset uses the complete reference and cannot be selected at decision time. All evaluated cases in each of the 17 target groups happen to be covered after expansion in the controlled-gap cohort, but those groups are not independent-climate replications.

We then propagated the same fixed source radii through the original masked archives and three rank constructions. Of 820 settings, 82 lack at least one annual distribution and 2 have no masked year; 736 nontrivial settings in 17 connected groups support both methods and all rank constructions. Two hypothetical quantile representations share the same expanded 5th and 95th percentiles: one changes only endpoints, and the other also tapers changes through interior quantiles. They preserve the same point median and use identical ranks, but their interpolated interior and normal-spacing tails differ. Table S19 shows empirical nominal 95% trend coverage, width and interval score. On the 685 archives with the earlier narrow source correction and all ranks, the independent-rank tapered D2 interval width is 0.403 with that earlier correction and 2.212 °C per decade with group maxima; the corresponding stochastic widths are 0.376 and 2.747. Matching the annual endpoints does not identify the trend distribution, and observed coverage on these selected masks does not certify coverage for future archives.

The independent annual audit reconstructed 88,053 source case–method predictions, 352 source group maxima, 14,714 target case–method scores and the annual summary cells. A separate calculation reconstructs 13,248 trend treatment rows from saved quantiles and rank arrays and checks all 685 overlapping earlier rank snapshots. These numerical audits use the same source and target observations and do not validate exchangeability, physical sensors, provider rights or an independent climate.

Table S10. Additional observation-only witnesses from approved USGS daily means. Every candidate remains in the qualification table. Sparse complete-reference support precludes an incidence or transfer inference.

Station	Year	Complete n	Observed L	Complete max	Excess
08377900	2016	3	14.990	14.853	0.137
08377900	2018	3	15.537	14.853	0.683
08377900	2020	3	15.737	14.853	0.883
08405200	2013	2	28.693	28.327	0.367
08405200	2015	2	28.703	28.327	0.377
08405200	2016	2	28.537	28.327	0.210
08405200	2019	2	28.833	28.327	0.507
08405200	2020	2	28.970	28.327	0.643
07364133	2015	1	29.427	28.767	0.660

Table S11. Per-archive matched numerical output. Annual errors exceed 0.5 °C. Counts and RMSE are original three-output rule / score-only comparator; RMSE uses common converged release-plus-bound working-likelihood fits and units °C per decade.

Scenario	Numeric	Errors	Bounds	Fits	RMSE
Calendar, offline	6,345	11 / 9	590 / 590	840	0.0198 / 0.0237
Calendar, forward	3,435	43 / 32	371 / 370	819	0.1240 / 0.1220
Controlled, offline	5,543	242 / 229	1378 / 1378	840	0.0405 / 0.0407
Controlled, forward	3,174	397 / 370	603 / 603	820	0.2024 / 0.1959

Table S12. Expanded paired controlled-gap stresses, original / stress. RMSE and interval widths are °C per decade. Coverage is for nominal 95% trend intervals; scores are also in °C per decade. Common fitted cohorts differ between estimators and scenarios.

Scenario	Estimator	Cases	RMSE	Coverage	Width	Score
Donor gaps	D2	205	0.1555 / 0.2024	92.2 / 86.8	0.465 / 0.479	0.756 / 1.039
Donor gaps	Stochastic	190	0.1386 / 0.1877	87.4 / 76.3	0.404 / 0.418	0.808 / 1.257
Heat, fixed years	D2	205	0.1555 / 0.1655	92.2 / 88.8	0.465 / 0.456	0.756 / 0.806
Heat, fixed years	Stochastic	187	0.1370 / 0.1552	87.7 / 87.2	0.403 / 0.383	0.776 / 0.884

Table S13. Source-model sensitivity on six-way common archives, baseline / stress. RMSE, width and score use °C per decade; coverage is the nominal 95% trend interval percentage. Strict changes only the affected HUC17 source model; candidate changes source providers and assignments across all regions.

Scenario	Estimator	Source	Cases	RMSE	Coverage	Width	Score
Donor gaps	D2	Original	205	0.1555 / 0.2024	92.2 / 86.8	0.465 / 0.479	0.756 / 1.039
Donor gaps	D2	Strict	205	0.1561 / 0.2027	92.2 / 86.8	0.467 / 0.475	0.758 / 1.050
Donor gaps	D2	Candidate	205	0.1514 / 0.2068	88.8 / 82.4	0.425 / 0.502	0.738 / 1.117
Donor gaps	Stochastic	Original	190	0.1386 / 0.1877	87.4 / 76.3	0.404 / 0.418	0.808 / 1.257
Donor gaps	Stochastic	Strict	190	0.1386 / 0.1877	87.4 / 76.3	0.404 / 0.418	0.808 / 1.257
Donor gaps	Stochastic	Candidate	190	0.1455 / 0.1969	86.8 / 76.3	0.392 / 0.461	0.799 / 1.364
Heat, fixed years	D2	Original	205	0.1555 / 0.1655	92.2 / 88.8	0.465 / 0.456	0.756 / 0.806
Heat, fixed years	D2	Strict	205	0.1561 / 0.1661	92.2 / 88.3	0.467 / 0.457	0.758 / 0.812
Heat, fixed years	D2	Candidate	205	0.1514 / 0.1642	88.8 / 86.8	0.425 / 0.410	0.738 / 0.877
Heat, fixed years	Stochastic	Original	187	0.1370 / 0.1552	87.7 / 87.2	0.403 / 0.383	0.776 / 0.884
Heat, fixed years	Stochastic	Strict	187	0.1370 / 0.1552	87.7 / 87.2	0.403 / 0.383	0.776 / 0.884
Heat, fixed years	Stochastic	Candidate	187	0.1443 / 0.1677	87.7 / 82.9	0.390 / 0.373	0.764 / 0.947

Table S14. Forward annual failure comparison, old raw score / projected-point score. An error exceeds 0.5 °C. Source-threshold counts can differ; the target count match is transductive and uses no target outcomes.

Mask	Selection	Cases	Numerical	Failures	Failure (%)
Calendar	Source threshold	3,806	3,431 / 3,112	6 / 6	0.17 / 0.19
Calendar	Target count match	3,806	3,431 / 3,431	6 / 6	0.17 / 0.17
Controlled	Source threshold	3,777	3,385 / 3,222	363 / 306	10.72 / 9.50
Controlled	Target count match	3,777	3,385 / 3,385	363 / 383	10.72 / 11.31

Table S15. Study screening rules on 2,656 eligible station-years in the 95-station candidate subset. Rejected-with-bound means the rejected incomplete year contains at least one complete observed 30-day window.

Rule	Accepted	Rejected	Unknown	Rejected with bound	Stations
Complete year	2,122	534	0	518	70
Annual 95%	2,550	106	0	90	48
Station warm season 90%	2,590	66	0	50	30
Complete July–August	2,472	184	0	168	58

Table S16. Post-hoc seasonal-window influence by existing project label and after removing two selected sites. "Above" is greater than 5×10^{-6} °C; the second count exceeds 0.5 °C. Labels are observational metadata, not independently verified treatments.

Cohort	Stations	Years	Above	> 0.5	Mean (°C)
All stations	95	2,122	779	304	0.255
Exclude two selected sites	93	2,086	749	279	0.197
Label: regulated	23	397	173	103	0.725
Label: unmatched gages	61	1,542	569	196	0.158
Label: unknown	11	183	37	5	0.049

Table S17. Retrospective screening in the 95-station candidate subset. "Above prior" counts rejected years whose observed 30-day window exceeds all earlier accepted reported windows. Witness column gives accepted / rejected / unknown among 35 previously selected incomplete years.

Rule	Accepted	Rejected	Unknown	Rejected with bound	Above prior	Witnesses
Complete year	2,122	534	0	518	86	0 / 35 / 0
Annual 95%	2,550	106	0	90	11	31 / 4 / 0
Warm 90%, leave year out	2,590	66	0	50	9	32 / 3 / 0
Warm 90%, past data	2,589	67	0	51	9	32 / 3 / 0
Warm 90%, past + 4 complete	2,487	42	127	31	3	27 / 2 / 6

Table S18. Nominal 90% annual intervals in common forward controlled-gap cases, baseline / source-group-maximum expansion. Width and interval score are °C; lower score is better.

Method	Stratum	Cases	Coverage (%)	Width	Score
D2	All	3,683	89.2 / 100.0	0.734 / 4.690	1.442 / 4.690
D2	Hidden peak	2,146	81.9 / 100.0	1.161 / 5.429	2.349 / 5.429
Stochastic	All	3,683	87.3 / 100.0	0.650 / 5.793	1.476 / 5.793
Stochastic	Hidden peak	2,146	78.2 / 100.0	1.078 / 6.525	2.496 / 6.525

Table S19. Nominal 95% forward trend intervals on 736 common nontrivial archives. Each entry is baseline / endpoint-only / tapered; widths and interval scores are °C per decade.

Method	Ranks	Coverage (%)	Width	Score
D2	Independent	88.6 / 100.0 / 100.0	0.399 / 3.102 / 2.222	0.735 / 3.102 / 2.222
D2	Shared parameter	88.7 / 100.0 / 100.0	0.395 / 3.035 / 2.178	0.732 / 3.035 / 2.178
D2	Residual blocks	88.9 / 100.0 / 100.0	0.400 / 3.022 / 2.229	0.714 / 3.022 / 2.229
Stochastic	Independent	84.5 / 100.0 / 100.0	0.366 / 3.968 / 2.751	0.896 / 3.968 / 2.751
Stochastic	Shared parameter	83.4 / 100.0 / 100.0	0.362 / 3.894 / 2.700	0.902 / 3.894 / 2.700
Stochastic	Residual blocks	83.8 / 100.0 / 100.0	0.369 / 3.881 / 2.781	0.882 / 3.881 / 2.781

S12 Annual support and hidden-peak diagnostics

The projection-stage and peak-visibility tables use common finite cases. The peak-visibility stratum depends on the withheld complete reference and is unavailable for an operational decision. Repeated masks can place the same physical station-year in both strata.

Table S20. Replay of annual interval stages on common finite-bound cases. Raw is the original model output; stochastic raw quantiles use a finite set of imputations that retain observed days. First applies the first explicit bound projection; Pre-final applies the fixed projection-aware offsets and sorts; Final applies the bound again. New is the count first becoming zero-width at the final projection. Zero widths use 10^{-8} °C tolerance. These are repeated mask cases, and no alternative calibration was fitted.

Family	Setting	Estimator	n	Raw	First	Pre-final	Final	New
calendar	forward	D2	3674	0	2	1096	1476	380
calendar	forward	Stochastic	3674	3090	3090	2977	2977	0
calendar	offline	D2	6919	0	0	4402	4538	136
calendar	offline	Stochastic	6919	5818	5818	5482	5482	0
crossed	forward	D2	3683	0	0	227	323	96
crossed	forward	Stochastic	3683	1362	1362	1091	1091	0
crossed	offline	D2	6919	0	1	1376	1391	15
crossed	offline	Stochastic	6919	2491	2491	2030	2030	0

Table S21. Retrospective annual performance on the two-estimator common finite-bound cohort, stratified using the complete reference. Match means that the observed lower bound equals the complete reported H30 within 10^{-8} °C; excess means the reference exceeds that bound. This outcome-defined stratum is unavailable at deployment. A station-year can occur in both strata under different masks. Intervals in brackets resample connected groups with 2,000 repetitions and are descriptive under the sampled archive, not independent-climate guarantees.

Family	Setting	Stratum	Method	n	Groups	MAE	Coverage	Group interval
calendar	forward	Match	D2	3125	17	0.000	100.0%	[100.0, 100.0]
calendar	forward	Match	Stochastic	3125	17	0.000	99.9%	[99.6, 100.0]
calendar	forward	Excess	D2	549	15	0.191	88.3%	[84.6, 93.2]
calendar	forward	Excess	Stochastic	549	15	0.151	87.8%	[84.1, 93.5]
calendar	offline	Match	D2	5891	18	0.000	99.9%	[99.6, 100.0]
calendar	offline	Match	Stochastic	5891	18	0.000	99.9%	[99.8, 100.0]
calendar	offline	Excess	D2	1028	16	0.122	90.6%	[86.8, 92.9]
calendar	offline	Excess	Stochastic	1028	16	0.120	83.1%	[79.8, 88.7]
crossed	forward	Match	D2	1537	17	0.012	99.3%	[98.9, 99.8]
crossed	forward	Match	Stochastic	1537	17	0.025	94.4%	[89.1, 99.3]
crossed	forward	Excess	D2	2146	17	0.378	83.2%	[72.6, 93.0]
crossed	forward	Excess	Stochastic	2146	17	0.345	81.6%	[72.2, 91.7]
crossed	offline	Match	D2	2787	18	0.008	99.3%	[98.9, 99.6]
crossed	offline	Match	Stochastic	2787	18	0.005	99.7%	[99.5, 99.9]
crossed	offline	Excess	D2	4132	18	0.285	90.7%	[88.5, 93.6]
crossed	offline	Excess	Stochastic	4132	18	0.289	87.9%	[84.6, 91.3]

Table S22. Exploratory source-only annual interval expansion on two-method common cases. Supported cases have at least three disjoint source groups and a positive empirical correction; unavailable and zero-correction regions remain in the denominator. Before and after use exactly the same supported cases. The source-group-weighted empirical correction has no finite-sample or climate-transfer guarantee; it was explored after inspecting the principal results. Coverage is percent, width and interval score are in °C.

Family	Setting	Method	Eligible	Supported	Coverage	Width	Interval score
calendar	forward	D2	3676	0	–	–	–
calendar	forward	Stochastic	3676	0	–	–	–
calendar	offline	D2	6926	0	–	–	–
calendar	offline	Stochastic	6926	0	–	–	–
crossed	forward	D2	3683	92	100.0→100.0	0.532→0.543	0.532→0.543
crossed	forward	Stochastic	3683	92	100.0→100.0	0.555→0.567	0.555→0.567
crossed	offline	D2	6919	0	–	–	–
crossed	offline	Stochastic	6919	0	–	–	–

S13 Source role audit for partitioned forward models

We re-derived the seed-zero bootstrap weights and checked them against the roles saved in 98 model bundles for the four evaluated source splits. The selected held test stations and their recorded connected groups did not enter any source role. However, in the before-2010 split, the role function defines held groups from test-period cases; training-only groups in the same named region may therefore remain in the source. The table reports positive-weight same-region source cases, not held-target observations. This is not a uniformly held-region design; the distinction affects the interpretation of forward transfer. It does not establish leakage of target-station hidden values or physical independence across groups.

S14 Post-hoc strict-region sensitivity

We refitted the two affected before-2010 seed-zero models after removing every source connected group that appears in the named target region. The original assignment and bootstrap multiplicities of other groups were retained; no replacement groups were drawn. Source interval and release thresholds were recalculated from the filtered roles. In the whole-archive experiment, only the three affected HUC17 stations were reanalysed, on the same 120 original masks; all other original results were retained. Each strict run checked the original mask and the absence of hidden target values. Both estimators were compared on the same station–repetition cases available in both versions. For forward controlled gaps this common cohort comprises 738 cases at 41 stations. Original and strict D2 trend RMSE are 0.1638 and 0.1645; the corresponding stochastic RMSE values are 0.1466 and 0.1466 °C per decade.

The future-year release analysis replaces predictions in both affected regions while preserving all target cases and the other regional models. Across 27,846 forward controlled-gap cases, the three-output rule publishes 63.1% and 64.5% in the original and strict comparisons, with 10.9% and 11.3% observed failures among the published values. In HUC17, publication changes from 62.7% to 71.6%, and the selected error rate from 6.0% to 8.2%. The selected sets differ, so the conditional rates cannot be interpreted without retention. The strict refits use fewer source observations; these paired comparisons do not identify the causal effect of an individual

group and do not establish external validity or safety. They do not recompute every analysis in this paper under strict regional exclusion. Case counts contain repeated masks and are not independent climates.

The source-role comparison additionally checked held test station identities across all source regions, the positive bootstrap weights and the effective split of classifier and probability-calibration groups. Only the two before-2010 models changed positive source weights; zero-weight role membership still matters for downstream group-list use, so this does not prove every downstream result unchanged.

The release-then-trend pipeline was replayed for all 120 HUC17 settings with the original masks, source-specific strict limits and newly fitted model. Original annual states reproduced for 1,440 station–year records. Across the affected records, 39 observed-bound outputs become numerical estimates; the other observed-bound and estimate states remain unchanged. The original and strict pooled controlled-gap comparisons comprise 820 common station–repetition cases and give released-values-plus-bounds working-likelihood trend RMSE 0.2024 and 0.1986 °C per decade, respectively, with mean working interval widths 3.26 and 3.29. These working intervals include regression variation and do not establish calibrated missingness-only coverage.

In the outcome-defined stratum with an unobserved peak, the 2,146 common forward controlled-gap annual cases have D2 nominal 90% coverage 83.2% originally and 82.9% under the stricter source. This retrospective stratum is unavailable to a decision maker and repeated masks are not independent climate realizations. The affected HUC17 source rule-calibration pool contains at most one eligible connected group before model prediction, below the minimum of three for both annual expansion and task calibration; strict refitting cannot change that source-group upper bound, although the corresponding radius and errors were not recalculated.

For forward controlled gaps, the same 818 station–repetition cases from 17 connected groups have every annual, tail and rank-coupling output in both source versions. Under constant tails, nominal 95% D2 trend coverage is 83.0% originally and 83.3% strictly; normal-spacing tails yield 88.8% and 88.9%, with widths 0.451 and 0.454 and interval scores 0.860 and 0.862. In the separate normal-tailed rank experiment, independent ranks yield 88.0% and 87.9% coverage and residual blocks

yield 89.0% and 89.0%. Original tail results reproduced on all 480 affected tail rows. These comparisons use the same random ranks within each tail or coupling experiment, but the tail and coupling experiments use different draws; they do not identify additive causes of undercoverage. Descriptive connected-group bootstrap intervals for strict-minus-original metrics accompany the aggregate results.

For the forward controlled-gap D2 versus stochastic comparison, 738 cases at 17 connected groups are available under both source versions and both estimators. The pooled D2-minus-stochastic MSE difference is +0.005350 originally and +0.005559 strictly; the descriptive connected-group bootstrap interval for the strict result is [+0.000957, +0.010431] (°C/decade)^2. The same resampled groups are used for both source versions. Removing Bavaria from these fixed predictions leaves 331 cases in 9 groups and changes the difference from -0.000143 originally to +0.000323 strictly. For empirical calendars the corresponding original and strict differences are -0.000213 and -0.000159 on 332 cases. These small changes can alter the sign of the regional comparison; the leave-one-region-out exercise does not refit source models and does not establish independent climates or significance.

The small controlled-gap stress comparison reanalyses 20 settings for one HUC17 station while retaining other stations' original results. Among 30 common baseline/stress station–repetition cases, original and strict baseline D2 RMSE are 0.0794 and 0.0803; for synchronized donor gaps they are 0.1164 and 0.1159, and for heat-associated gaps 0.0885 and 0.0902 °C per decade. Original scenario metrics and masks reproduce. The heat-associated scenario changes both the identity and potentially the number of masked years, so these are exploratory paired scenario comparisons, not isolated causal effects or independent climate tests.

A separate recalculation of the transductive, same-number numerical-output comparator under the strict source model publishes 17,961 controlled-gap values: three-output and score-only failure rates are 11.26% and 7.15%, respectively. The score threshold uses the target feature distribution and the three-output count; counts and selected cases differ between source versions, and lower-bound information is not held equal.

Table S23. Seed-zero before-2010 model roles with positive-weight source cases in the named target region. Bootstrap-weighted counts include repeated role draws and do not represent independent samples.

Region	Role	Groups	Stations	Cases	Weighted cases
nve_hydapi	interval_calibration	2	3	431	695
usgs_huc17	estimator	1	1	140	280

S15 Source-provider dependence of whole-archive models

The 42 whole-archive target stations belong to providers with recorded reuse paths, but source-model training has additional provenance. Recomputing the source role assignments and checking them against all 13 saved seed-zero model bundles shows that 13 estimator roles and 6 interval-calibration roles have positive-weight LUBW source cases. Source cases are repeated masks; their counts are not independent observations. Positive source weights establish model dependence but do not resolve the legal status of any particular derivative. Individual source, model, target, mask and reviewer-access rights still require confirmation. A post-hoc candidate-provider strict-region refit is reported separately and does not replace the main results.

S16 Candidate-provider, strict-region source sensitivity

After observing the original results and LUBW source dependence, we refitted all 13 whole-archive source models on seven candidate providers with recorded reuse paths. Role assignments and bootstrap draws were recomputed using the filtered data, and every group occurring in the target region was excluded from source roles. An independent role recount found no positive-weight target-region source case or post-2009 forward source case. Original target masks and daily features were reused. All 3,320 archive settings, masks and the unchanged plug-in baselines were checked. Removing providers, changing source assignments and strict regional exclusion jointly change the analysis; the result is not a causal attribution to one of them. The filtered inputs remain processed study data, not provider-native raw records. This after-the-fact sensitivity neither establishes legal redistribution rights nor independent climate transfer, and it does not replace the original analysis. A descriptive connected-group bootstrap resamples the same groups for both source versions and both estimators. It does not establish independent climates or a population transfer guarantee. The corresponding paired MSE differences and bootstrap intervals are supplied with the candidate-source summary.

Table S24. LUBW cases with positive source-role weights, not independent station counts. Groups with bootstrap multiplicity zero do not contribute.

Mode	Region	Estimator case rows	Interval calibration case rows
forward	foen	383	0
forward	gkd_bayern	287	0
forward	usgs_huc03	383	0
forward	usgs_huc04	96	0
forward	usgs_huc10	191	96
forward	usgs_huc17	191	0
offline	foen	288	311
offline	gkd_bayern	862	2732
offline	syke	1030	311
offline	usgs_huc03	3475	0
offline	usgs_huc04	1461	168
offline	usgs_huc10	1150	0
offline	usgs_huc17	1246	2732

Table S25. Trend RMSE in °C per decade on the four-way common cases: both methods and both source-model versions must be available. Repetitions are not independent climates.

Family	Mode	Cases	Groups	D2 orig.	Class. orig.	D2 cand.	Class. cand.
calendar	forward	701	17	0.06178	0.05134	0.06033	0.05822
calendar	offline	835	18	0.00873	0.01124	0.00890	0.01119
crossed	forward	738	17	0.16381	0.14658	0.15806	0.15600
crossed	offline	838	18	0.02780	0.03208	0.02868	0.03195

Table S26. Nominal 95% D2 trend-interval results on cases where both versions are available. Width is in °C per decade; this comparison is descriptive and the selected cases may differ from the four-way point cohort.

Family	Mode	Cases	Cover orig.	Cover cand.	Width orig.	Width cand.
calendar	forward	820	96.0%	95.7%	0.116	0.115
calendar	offline	840	95.0%	95.2%	0.018	0.018
crossed	forward	820	88.0%	87.6%	0.450	0.408
crossed	offline	840	94.0%	92.9%	0.089	0.086

Table S27. Symmetric MSE decomposition on cases with all four retention-by-estimator cells available under both versions. Fractions divide the mean retention contribution by mean total improvement; they are not causal or universal.

Family	Mode	Estimator	Cases	Groups	Original fraction	Candidate fraction
calendar	forward	D2	820	17	0.799	0.783
calendar	forward	classical	701	17	0.744	0.767
calendar	offline	D2	840	18	0.887	0.891
calendar	offline	classical	835	18	0.931	0.930
crossed	forward	D2	807	17	0.939	0.937
crossed	forward	classical	725	17	0.925	0.931
crossed	offline	D2	840	18	0.930	0.933
crossed	offline	classical	838	18	0.942	0.942

S17 Annual uncertainty under candidate-source refits

The follow-up compares the same target cases where both methods and both source versions have finite seven-quantile distributions and a finite observed bound. The original 24 annual summary cells were independently reproduced on their original cohorts. Target identities, bounds and connected groups agree between versions. The decomposition by whether the complete reference exceeds the visible bound uses the withheld reference and is unavailable at decision time. Candidate-versus-original differences are post hoc; changed source providers, role assignments and strict region exclusion cannot be separated. The quoted CRPS uses the stated normal-spacing tail continuation, and no new calibration or independent climate validation is established. For forward controlled gaps with hidden peaks, the paired candidate-minus-original classical coverage difference is -3.4 percentage points with descriptive connected-group bootstrap interval [-8.6, 2.0]. Removing FOEN or Bavaria from the fixed evaluation cases changes that difference to +0.3 or -8.9 percentage points, respectively. Regions were not refitted in this deletion, and connected groups are not proven independent climates.

S18 Single-year source expansion for annual intervals

After reviewing prior annual and trend results, we used single-year masked source cases from the candidate models' rule-calibration groups. These groups do not enter the estimators, original interval calibration or risk-learning roles; every target-region group is excluded. Some groups contributed to the separate risk-rule component; the pool is not an unopened independent dataset. Forward source cases are from before 2010. For each family, mode, region and estimator we estimate an equal-group-weighted empirical 90th percentile of nonnegative interval misses. A correction is applied only with at least three available source groups and a positive radius. This rule is exploratory, and the source cases hide a single year rather than an entire degraded archive. It neither establishes group exchangeability nor identifies a calibrated joint trend distribution. At most eight forward groups are assigned to this role, below the nine needed for a finite one-maximum-per-group 90% rank; the group-maximum diagnostic was not applied. Re-prediction of the 52 source support cells, verification of individual target intervals and a separate two-method support-intersection check passed. Group-resampling and leave-region-out summaries are

Table S28. Annual 90% interval comparison on the same four-way common cases. Hidden peak means that the complete reported H30 exceeds the observed bound; this is an outcome-defined stratum. Widths are in °C. Repeated masks are not independent climates.

Family / mode	Stratum	Method	Cases	Groups	Cover old	Cover new	Width old	Width new
calendar / forward	all	D2	3674	17	98.3%	97.9%	0.118	0.135
calendar / forward	all	classical	3674	17	98.1%	97.9%	0.088	0.093
calendar / forward	hidden	D2	549	15	88.3%	86.3%	0.690	0.680
calendar / forward	hidden	classical	549	15	87.8%	86.7%	0.566	0.610
calendar / offline	all	D2	6919	18	98.5%	98.5%	0.095	0.093
calendar / offline	all	classical	6919	18	97.4%	97.7%	0.066	0.069
calendar / offline	hidden	D2	1028	16	90.6%	90.8%	0.573	0.578
calendar / offline	hidden	classical	1028	16	83.1%	85.1%	0.427	0.434
crossed / forward	all	D2	3683	17	90.0%	89.2%	0.826	0.734
crossed / forward	all	classical	3683	17	87.0%	87.3%	0.693	0.650
crossed / forward	hidden	D2	2146	17	83.2%	81.9%	1.299	1.161
crossed / forward	hidden	classical	2146	17	81.6%	78.2%	1.121	1.078
crossed / offline	all	D2	6919	18	94.1%	92.8%	0.730	0.701
crossed / offline	all	classical	6919	18	92.7%	92.4%	0.660	0.639
crossed / offline	hidden	D2	4132	18	90.7%	88.6%	1.171	1.127
crossed / offline	hidden	classical	4132	18	87.9%	87.4%	1.068	1.037

Table S29. Exploratory interval expansion: same target cases must be supported for both methods. Baseline and expanded scores use the identical supported subset; all-case baselines are retained separately in the supplied CSV. Hidden-peak strata depend on the complete reference. Widths are in °C.

Family / mode	Stratum	Method	Eligible	Supported	Groups	Cover before	Cover after	Width before	Width after
calendar / forward	all	D2	3674	0	0	–	–	–	–
calendar / forward	all	classical	3674	0	0	–	–	–	–
calendar / forward	hidden	D2	549	0	0	–	–	–	–
calendar / forward	hidden	classical	549	0	0	–	–	–	–
calendar / offline	all	D2	6919	0	0	–	–	–	–
calendar / offline	all	classical	6919	0	0	–	–	–	–
calendar / offline	hidden	D2	1028	0	0	–	–	–	–
calendar / offline	hidden	classical	1028	0	0	–	–	–	–
crossed / forward	all	D2	3683	3411	14	88.7%	89.9%	0.705	0.755
crossed / forward	all	classical	3683	3411	14	86.6%	89.1%	0.608	0.696
crossed / forward	hidden	D2	2146	1982	14	81.0%	82.9%	1.110	1.171
crossed / forward	hidden	classical	2146	1982	14	76.9%	81.3%	1.009	1.115
crossed / offline	all	D2	6919	0	0	–	–	–	–
crossed / offline	all	classical	6919	0	0	–	–	–	–
crossed / offline	hidden	D2	4132	0	0	–	–	–	–
crossed / offline	hidden	classical	4132	0	0	–	–	–	–

supplied; they do not prove independent climates. Unsupported cells and unchanged all-case baselines remain visible. The source group-mass empirical percentile is not a finite-sample coverage guarantee.

S19 Trend sensitivity to annual interval expansion

After examining the source-only annual correction and the earlier trend results, we propagated it through the original 256-draw normal-spacing quantile representation. The endpoint-only scheme moves the 5th and 95th quantiles while holding the five interior quantiles fixed. The tapered scheme also shifts the 10th, 25th, 75th and 90th quantiles by 0.8, 0.4, 0.4 and 0.8 times the radius, respectively; the median and complete-year values are unchanged. Both schemes use the same adjusted 90% endpoints and respect observed lower bounds. Their interior and normal-spacing outer tails differ. The same independent, auxiliary shared-parameter and residual-block rank matrices are reused within each scheme; no calibrated joint distribution is implied. Among 820 forward controlled-gap station–repetition settings, 760 belong to regions with source support for both methods. Seventy-three of these lack at least one annual distribution, and two have no incomplete year, leaving 685 common nontrivial settings. Original saved trend intervals are reproduced before adjustment. The target selection and calibration remain retrospective. Connected-group paired bootstrap and fixed-prediction leave-one-region-out summaries are supplied. They are descriptive, use only 14 target groups and do not establish independent climates. Annual 90% endpoint correction does not identify the inner or outer distribution, cure annual undercoverage, guarantee trend coverage, or resolve source rights. The underlying case records and rank arrays remain local pending permission.

Table S30. Nominal 95% finite-record trend interval comparison on the same 685 station–repetition cases, 38 stations and 14 connected groups. Width and interval score are in °C per decade; repeated masks are not independent climates.

Method	Rank construction	Marginal scheme	Coverage	Width	Interval score
D2	independent	baseline	87.9%	0.382	0.742
D2	independent	endpoints	89.3%	0.409	0.736
D2	independent	tapered	88.9%	0.403	0.735
D2	shared parameter	baseline	88.0%	0.379	0.741
D2	shared parameter	endpoints	89.2%	0.405	0.730
D2	shared parameter	tapered	89.2%	0.399	0.733
D2	residual blocks	baseline	88.2%	0.385	0.721
D2	residual blocks	endpoints	89.8%	0.411	0.708
D2	residual blocks	tapered	89.5%	0.406	0.707
classical	independent	baseline	83.4%	0.344	0.912
classical	independent	endpoints	86.9%	0.387	0.856
classical	independent	tapered	85.5%	0.376	0.866
classical	shared parameter	baseline	82.2%	0.342	0.922
classical	shared parameter	endpoints	86.3%	0.383	0.863
classical	shared parameter	tapered	85.0%	0.373	0.872
classical	residual blocks	baseline	82.6%	0.349	0.901
classical	residual blocks	endpoints	86.4%	0.391	0.845
classical	residual blocks	tapered	85.7%	0.381	0.853

S20 External reference qualification

The Greater Wellington intake returned records for nine selected sites. Under the originally specified quality-code-600 criterion no complete reference hot year was qualified; the provider describes code 200 as unverified, while other returned legacy code meanings remain unresolved. A NEON PRT archive supplied half-hourly records at 24 river or stream sites. With one fixed location code per site, no site supplied four complete years under either 48 quality-passing intervals per UTC day or the 46-of-48 sensitivity. The latter is an observed-interval average rather than a complete physical day; a location code does not establish unchanged hardware.

After the original intake and quality screen, we separately tested an optimistic time-support bound on the same nine cached Greater Wellington records: all quality flags were ignored, while the original fixed-NZST cadence and fully observed day rule were retained and finite values required. Across these records there were 10 potentially complete July–June years, with zero to two per site; no site reached four. This post-hoc feasibility screen is conditional on the cached series and these completeness rules, not a quality assessment, reference qualification, or recovery score. The annual counts were recomputed independently from the same cached XML using a separate day-count implementation; the XML parser and source observations were shared.

An initial 15-station USGS intake had already been processed in the national panel, and ten stations had been scored in an earlier external comparison. The subsequent eight-station roster excluded these and other identified source, evaluation and cached-value contacts; no station passed the four-year reference requirement after the documented quality checks. A further five metadata candidates with prior network screening had no four years of complete approved daily counts in 2010–2024; counts can include estimated values and do not establish subdaily quality. Five selected Réunion Hub'Eau sites provided records with quality code 1, but no site supplied four complete July–June years under the 24-distinct-valid-hour rule; the provider-day time zone is unresolved. The same intake results do not describe the fraction of eligible stations in any complete network.

At ECCC Red River at Emerson, the USGS alias and an upstream source station place the record on a previously encountered river. The files use the word

VALIDATED and, for some rows, a station identifier different from the catalog; the mapping to literal V and numeric quality grades and the daylight-saving convention are unresolved. Under the fixed literal V and grade 1/2 rule, zero complete local-clock years qualified. These exclusions are not an assertion that the rejected observations are erroneous; no numerical recovery score was calculated for this site. Data-source citations appear in the main article.

S21 Prior Australian reference exposure

An earlier Australian recruitment stage identified 80 observation-supported candidate station-years from 52 stations. The historical intake record states that its designated F2 formal test was closed; this audit does not run that model or calculate new F2 errors. Subsequently, a separate, earlier-method Australian evaluation used calendar-year cases from the same acquired source archive. All 80 of the candidate station-years appear in that case file; 79 had at least one case in the old evaluation's eligible case set, including all 10 of the 10 originally designated F2 test-role station-years. The prior evaluation contained 30,728 eligible masked cases across its wider roster. These are previously evaluated records for a different pipeline, not unopened target station-years for a new independent test.

A later Australian development analysis used July–June rather than calendar-year windows. Counting physical time overlap rather than matching the year labels alone, 74 of the original 80 calendar station-years intersect a saved development window, of which 73 intersect at least one eligible development case. A separate Arrow/Python recount reproduced these counts. Matching station and interval does not prove that masked days are identical, that the F2 model was evaluated, or that data leaked into source-model training. The original native-sampling criterion allowed 20 observed hours per day and does not establish a complete 24-hour physical reference. These previously opened Australian records cannot supply a fresh independent confirmation of the current pipeline.

S22 Exploratory British Columbia daily-reference comparison

All 23 author-selected published stations were retained for qualification. Four stations each have four complete calendar years in the published daily series. The original study accepted daily means based on at least 20 hourly values after re-

searcher quality checks; hourly counts, original flags and the day convention are not available in this derivative. The source-water cleaning used water–air comparisons. A complete published year is not equivalent to 24 independently quality-checked hours per day. The 23-station author cohort was selected for data availability and these results are unregistered. The fixed global source model and independent source-role recalibration are used; local reconstruction models are refit after holding out the scored year. Each year is masked separately, with 30-, 60- and 90-day controlled gaps. NASA POWER air forcing uses coordinates rounded to 0.01 degree from the source article; there are no same-panel donor stations. The score treats the published daily mean as reference, not as error-free physical truth. Repeated gaps in a station-year and the four stations are dependent; three stations lie in the Fraser River basin (Nechako, Thompson and Fraser). These are descriptive results, without population uncertainty or a natural-outage validation. Across 372 masked

Table S31. Controlled-gap annual H30 mean absolute errors in degrees Celsius. Each row has four published-daily reference years; mask counts are not independent observations.

Station	Masks	Plug-in	D2	Stochastic
08FF003	72	0.833	0.475	0.520
08JA017	72	0.241	0.243	0.203
08LF051	72	0.383	0.207	0.220
08MF040	72	0.605	0.298	0.265

cases, independently recomputed 30-day reference maxima, observed-window bounds, missing counts, and 52 point and interval summary metrics agree with stored results within numerical tolerance. As a limited noninterference probe, one 90-day case per station was regenerated after adding 5 degrees C to hidden target values. All 4 probes retained the same decision-time model features and numerical predictions, while the evaluation reference changed. These probes do not establish noninterference for every case, model path, or experiment. The attribution, candidate roster, year-level qualification, individual station summaries and code are included in the local evidence package. Prior-identifier and geographic proximity checks are best effort and cannot prove independence from every previously opened record.

S23 Nova Scotia verified workbooks and open-portal comparison

Four river locations were fixed from the provider's historical workbook list before opening their temperature sequences. The workbooks are labelled verified, but their inspected headers do not identify per-hour temperature quality or estimated-value flags; one header states UTC–04:00 and the remaining headers do not establish a time basis. We count complete reported clock-label years, requiring 24 unique hourly labels with finite numeric cells per calendar day, without claiming these labels correspond to 24 independent physical hours or nonestimated values. The openly licensed portal was accessed only after the workbook completeness screen; it is another release of the same monitoring programme, not an independent sensor. Of 14 station-years with complete labels in both releases, 12 have every numeric value equal within the originally specified 10^{-6} °C tolerance. These include at least four such years at 2 stations. Across 262,667 shared unique numeric labels, 12,810 differ at that tolerance; 1,086 exceed 0.5 °C in a post-hoc diagnostic, with a maximum 4.493 °C. These comparisons do not identify which version is correct, and clock-label matching does not resolve physical time. The original reference screen did not score a reconstruction method; a later, explicitly exploratory amendment is reported separately. The portal source is Government of Nova Scotia, dataset bkfi-mjgw, Open Government Licence – Nova Scotia; legacy workbook reuse remains unresolved.

S24 Nova Scotia reported-clock exploratory masking

The preceding qualification did not establish strict reference quality. After inspecting availability and two-release agreement, a separate unregistered exploratory amendment fixed two sites and eight station-years before any model prediction errors were computed. Each selected year has 24 finite reported hourly labels per day and agrees numerically across the two releases of the same monitoring programme. Time zone, estimated-value status and point-level quality remain unresolved. The reported daily value is the mean of the 24 reported hourly values; it is not guaranteed to represent a physical civil day or an error-free temperature. The fixed global source model and source-only interval calibration were reused. Each year was masked separately with 30-, 60- and 90-day controlled gaps, using the other three selected years for local history; no empirical-calendar family or

whole-archive degradation was evaluated. NASA POWER forcing uses local solar time and coordinates rounded to 0.01 degree. The adjacent-normal tail continuation is an assumption for CRPS. The 144 cases are repeated masks at two sites and eight station-years, not 144 independent watersheds. An independent arithmetic audit recomputed reference H30, bounds, masks and reported scores. In two post-hoc hidden-value probes, adding 5 °C to hidden days changed the true H30 but did not change decision features or predictions when identically serialized controls were compared. Those two probes are not an exhaustive leakage proof.

S25 Nova Scotia reported-day boundary sensitivity

After inspecting the exploratory results, we tested hypothetical daily bins shifted by −1, 0 and +1 hour relative to the recorded clock labels. We required all original eight station-years and the adjacent boundary hours to be unique, finite and equal within 10^{-6} °C across the two releases; missing or disputed hours were not filled. Daily water inputs and local leave-one-year-out models were rebuilt for each shift, with the original masks, global source model, source calibration and NASA POWER local-solar-time forcing held fixed. All 144 masks are available across all three shifts for both methods; the zero-shift replay agrees with the original outputs. Across the eight reported reference years the largest H30 change is 0.0023 °C, while a post-hoc casewise diagnostic finds a largest D2 point-prediction change of 0.094 °C. This does not identify the true timezone or explain the cause of prediction changes; hourly values, quality and meteorological day alignment remain uncertain. The two sites and repeated masks do not justify a population-level ranking or independent-climate inference.

Table S32. Retrospective 2009–2017 reported-label completeness and release concordance. Equal means that every hourly numeric value in the common complete-label year matches to 10^{-6} °C at the same reported clock label. Quality, estimated-value status, time basis and climate independence are not established.

Station	Workbook	Portal	Both	Fully equal
NS01DL0009	4	5	4	4
NS01ED9111	2	2	1	1
NS01EF0002	4	4	3	3
NS01FB0001	6	6	6	4

Table S33. Reported-clock exploratory controlled-gap scores on identical available cases. Coverage is percent, and MAE, width and interval score are in °C. These are two sites, not independent climatic replications; source quality and time basis remain unresolved.

Method	Cases	MAE	90% coverage	90% width	Interval score
D2	144	0.311	88.2	1.000	1.572
Stochastic	144	0.313	93.1	1.281	1.706

Table S34. Post-hoc descriptive paired results by gap length and site. Counts are repeated masks; group differences are not inferential tests.

Axis	Level	Cases	D2 MAE	Stochastic MAE	D2 coverage	Stochastic coverage
Gap days	30	48	0.247	0.231	89.6	93.8
Gap days	60	48	0.297	0.319	93.8	95.8
Gap days	90	48	0.387	0.388	81.2	89.6
Station	NS01DL0009	72	0.400	0.393	81.9	93.1
Station	NS01FB0001	72	0.221	0.232	94.4	93.1

Table S35. Paired exploratory reported-day boundary sensitivity with identical original masks. Coverage is percent and error, width and interval score are in °C. Local stages are refit; source model, source calibration and forcing are held fixed. Two sites and eight station-years are not independent climatic replications.

Shift (h)	Method	Cases	MAE	90% coverage	Width	Interval score
-1	D2	144	0.310	88.9	1.000	1.555
-1	Stochastic	144	0.313	92.4	1.283	1.704
0	D2	144	0.311	88.2	1.000	1.572
0	Stochastic	144	0.313	93.1	1.281	1.706
1	D2	144	0.315	86.8	0.998	1.572
1	Stochastic	144	0.313	93.1	1.278	1.731

S26 Independent observed-record arithmetic re-audits

Using the same local canonical float32 daily archive and an independent convolution implementation, we recomputed 2,139 complete reference station-years and 545 incomplete station-years at 96 eligible stations. All 35 witness identities, saved 30-day values and complete-year reference maxima agreed within the stated numerical tolerance. This is an arithmetic re-audit of the same processed archive, not an independent check of provider-native records or sensor error.

Comparing 971,673 finite values from 34 selected project input panels with the canonical archive yielded no missingness or float32-cast discrepancies. The largest absolute rounding difference is 0.00000153 degrees C; 0 witness identities and 0 classifications at the 0.5-degree excess threshold changed. The input panels are processed project files, not provider-native instrument records.

Across 2,700 eligible station-years, independently recalculated annual coverage, available 30-day maxima, July–August completeness and longest missing runs reproduced 210 witness–rule decisions. For each of 35 witness years, increasing its observed values by 5 degrees C did not change the leave-target-year-out warm-season climatology. The old fixed-season fraction remains sourced from saved metadata; these probes do not cover every year or other pipelines.

For the ten additional USGS sites, independent parsing of the saved daily API responses reproduced 8 approved-complete reference station-years and the 9 certified observed windows. The original quality-screened window selection also replays from the saved payloads, but the separate arithmetic audit does not independently implement that selection. Approved estimated values remain in complete-year references, but the certified windows contain no estimated days. These records do not establish subdaily completeness, physical measurement accuracy or independent recovery performance.

A separate candidate review subset contains processed daily values for 95 of 96 previously eligible source stations, including all 35 witnesses. One LUBW station is omitted pending permission. The subset verifies the included reported-series arithmetic and rule decisions, but cannot independently confirm the omitted station or the original full eligible-station denominator.